Math 21 Curriculum

Algebra and its Functions | Middlesex School Mathematics

COURSE AT A GLANCE

Review Content

REVIEW

Primary resources: Intermediate Algebra 2e 3.2 Slope of a Line; Intermediate Algebra 2e 3.1 Graph Linear Equations in Two Variables; College Algebra 2e 4.1 Linear Functions; Intermediate Algebra 2e 4.1 Solve Systems of Linear Equations with Two Variables; Intermediate Algebra 2e 5.2 Properties of Exponents and Scientific Notation

UNIT LEARNING OBJECTIVES

I can determine and interpret the slope of a line.

I can graph a line from an equation and write an equation from given information.

I can apply a linear function to a context.

I can identify the solution of a linear system as an intersection point.

I can solve a simple two-variable linear system using an appropriate method.

I can simplify an expression using basic positive-integer exponent rules.

Functions and Function Interpretation

REQUIRED

Primary resources: Intermediate Algebra 2e 3.5 Relations and Functions; College Algebra 2e 3.3 Rates of Change and Behavior of Graphs
Supplemental resources: Intermediate Algebra 2e 3.6 Graphs of Functions; College Algebra 2e 3.1 Functions and Function Notation; College Algebra 2e 4.1 Linear Functions

UNIT LEARNING OBJECTIVES

I can evaluate a function from a formula, graph, or table.

I can determine whether a relation is a function.

I can determine domain and range from a graph or table.

I can solve for an input that produces a specified output.

I can interpret inputs, outputs, domain, and range in context.

I can calculate and interpret average rate of change.

I can compare functions represented by formulas, graphs, or tables.

Polynomials and Factoring

REQUIRED

Primary resources: 5.1 Add and Subtract Polynomials; 5.3 Multiply Polynomials; 6.1 Greatest Common Factor and Factor by Grouping; 6.2 Factor Trinomials; 6.3 Factor Special Products; 6.5 Polynomial Equations

UNIT LEARNING OBJECTIVES

I can add and subtract polynomials.

I can multiply polynomials.

I can factor out a greatest common factor.

I can factor a quadratic trinomial.

I can factor a difference of squares and other required special cases.

I can use factoring to determine the zeros of a polynomial.

Rational Expressions and Equations

REQUIRED

Primary resources: 7.1 Multiply and Divide Rational Expressions; 7.2 Add and Subtract Rational Expressions; 7.4 Solve Rational Equations

UNIT LEARNING OBJECTIVES

I can state excluded values for a rational expression.

I can simplify a rational expression while preserving its restrictions.

I can multiply and divide rational expressions.

I can add and subtract rational expressions using a common denominator.

I can solve a rational equation and reject extraneous solutions.

Exponents and Radicals

REQUIRED

Primary resources: Intermediate Algebra 2e 8.3 Simplify Rational Exponents; Intermediate Algebra 2e 8.2 Simplify Radical Expressions; Intermediate Algebra 2e 8.6 Solve Radical Equations
Supplemental resources: College Algebra 2e 2.6 Other Types of Equations

UNIT LEARNING OBJECTIVES

I can simplify expressions containing negative exponents.

I can rewrite expressions between radical and rational-exponent forms.

I can simplify radical expressions.

I can apply exponent rules to expressions with integer and rational exponents.

I can solve a basic equation involving a radical or rational exponent and check for extraneous solutions.

Quadratic Functions and Equations

REQUIRED

Primary resources: 9.6 Graph Quadratic Functions Using Properties; 9.7 Graph Quadratic Functions Using Transformations; 6.5 Polynomial Equations; 9.1 Solve Quadratic Equations Using the Square Root Property; 9.2 Solve Quadratic Equations by Completing the Square; 9.3 Solve Quadratic Equations Using the Quadratic Formula; 9.5 Solve Applications of Quadratic Equations
Supplemental resources: 9.6 Graph Quadratic Functions Using Properties; 6.5 Polynomial Equations

UNIT LEARNING OBJECTIVES

I can identify key features of a quadratic function from its graph or equation.

I can connect standard, factored, and vertex forms to features of a parabola.

I can solve a quadratic equation by factoring.

I can solve a quadratic equation using the square-root property.

I can solve a quadratic equation by completing the square.

I can solve a quadratic equation using the quadratic formula.

I can choose an efficient method for solving a quadratic equation.

I can graph a quadratic function using its vertex, intercepts, and symmetry.

I can create and interpret a quadratic model for a projectile, area, or optimization problem.

Extension Content

EXTENSION

Primary resources: 8.8 Use the Complex Number System; 9.3 Solve Quadratic Equations Using the Quadratic Formula

UNIT LEARNING OBJECTIVES

I can perform basic arithmetic with complex numbers.

I can express the nonreal solutions of a quadratic equation using complex numbers.

IN-DEPTH CURRICULUM

SKILL PROGRESSION I Introduce R Reinforce D Deepen A Apply

Review Content

REVIEW

OpenStax coverage: **Primary resources:** Intermediate Algebra 2e 3.2 Slope of a Line; Intermediate Algebra 2e 3.1 Graph Linear Equations in Two Variables; College Algebra 2e 4.1 Linear Functions; Intermediate Algebra 2e 4.1 Solve Systems of Linear Equations with Two Variables; Intermediate Algebra 2e 5.2 Properties of Exponents and Scientific Notation

M21-REV-001

I can determine and interpret the slope of a line.

OpenStax: **Primary resources:** 3.2 Slope of a Line

SUPPORTING SKILLS

- R Determine slope from a graph.
  SK-LIN-SLOPE-GRAPH · first introduced in Math 12
- R Calculate slope from two points.
  SK-LIN-SLOPE-POINTS · first introduced in Math 12
- R Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

M21-REV-002

I can graph a line from an equation and write an equation from given information.

OpenStax: **Primary resources:** 3.1 Graph Linear Equations in Two Variables

SUPPORTING SKILLS

- R Convert among common forms of a linear equation.
  SK-LIN-FORM-CONVERT · first introduced in Math 12
- R Use slope to generate additional points on a line.
  SK-LIN-GRAPH-SLOPE · first introduced in Math 12
- R Plot an intercept accurately on a coordinate plane.
  SK-LIN-PLOT-INTERCEPT · first introduced in Math 12
- A Calculate slope from two points.
  SK-LIN-SLOPE-POINTS · first introduced in Math 12

M21-REV-003

I can apply a linear function to a context.

OpenStax: **Primary resources:** 4.1 Linear Functions

SUPPORTING SKILLS

- I Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET
- R Restrict a model to meaningful contextual inputs.
  SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- A Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

M21-REV-004

I can identify the solution of a linear system as an intersection point.

OpenStax: **Primary resources:** 4.1 Solve Systems of Linear Equations with Two Variables

SUPPORTING SKILLS

- R Locate and interpret the intersection of two lines.
  SK-SYS-GRAPH-INTERSECTION · first introduced in Math 12

M21-REV-005

I can solve a simple two-variable linear system using an appropriate method.

OpenStax: **Primary resources:** 4.1 Solve Systems of Linear Equations with Two Variables

SUPPORTING SKILLS

- R Check an ordered pair in both equations of a system.
  SK-SYS-CHECK · first introduced in Math 12
- R Add equations to eliminate a variable.
  SK-SYS-ELIMINATE · first introduced in Math 12
- R Locate and interpret the intersection of two lines.
  SK-SYS-GRAPH-INTERSECTION · first introduced in Math 12
- R Substitute an equivalent expression for a variable.
  SK-SYS-SUBSTITUTE · first introduced in Math 12

M21-REV-006

I can simplify an expression using basic positive-integer exponent rules.

OpenStax: **Primary resources:** 5.2 Properties of Exponents and Scientific Notation

SUPPORTING SKILLS

- R Apply the power rule for exponents.
  SK-EXP-POWER · first introduced in Math 12
- R Apply the product rule for exponents.
  SK-EXP-PRODUCT · first introduced in Math 12
- R Apply the quotient rule for exponents.
  SK-EXP-QUOTIENT · first introduced in Math 12

Functions and Function Interpretation REQUIRED

OpenStax coverage: **Primary resources:** Intermediate Algebra 2e 3.5 Relations and Functions; College Algebra 2e 3.3 Rates of Change and Behavior of Graphs
Supplemental resources: Intermediate Algebra 2e 3.6 Graphs of Functions; College Algebra 2e 3.1 Functions and Function Notation; College Algebra 2e 4.1 Linear Functions

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M21-FUN-001

I can evaluate a function from a formula, graph, or table.

OpenStax: **Primary resources:** 3.5 Relations and Functions
Supplemental resources: 3.6 Graphs of Functions

SUPPORTING SKILLS

- A Simplify an algebraic expression using operation properties.
  SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- D Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- A Identify the input and output represented by function notation.
  SK-FUNC-NOTATION-READ · first introduced in Math 12

M21-FUN-002

I can determine whether a relation is a function.

OpenStax: **Primary resources:** 3.5 Relations and Functions

SUPPORTING SKILLS

- D Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- D Apply the vertical line test to a graph.
  SK-FUNC-VERTICAL-LINE · first introduced in Math 12

M21-FUN-003

I can determine domain and range from a graph or table.

OpenStax: **Primary resources:** 3.5 Relations and Functions

SUPPORTING SKILLS

- D Read domain and range from a graph.
  SK-FUNC-DOMAIN-RANGE-GRAPH · first introduced in Math 12
- D Read domain and range from a table.
  SK-FUNC-DOMAIN-RANGE-TABLE · first introduced in Math 12

M21-FUN-004

I can solve for an input that produces a specified output.

OpenStax: **Primary resources:** 3.5 Relations and Functions

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I Solve for an input associated with a specified output.
  SK-FUNC-SOLVE-INPUT

M21-FUN-005

I can interpret inputs, outputs, domain, and range in context.

OpenStax: **Primary resources:** 3.5 Relations and Functions

SUPPORTING SKILLS

- D Restrict domain to values meaningful in a context.
  SK-FUNC-CONTEXT-DOMAIN · first introduced in Math 12
- D Interpret the range of a function in a context.
  SK-FUNC-CONTEXT-RANGE · first introduced in Math 12
- D Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- D Restrict a model to meaningful contextual inputs.
  SK-MODEL-VALID-DOMAIN · first introduced in Math 12

M21-FUN-006

I can calculate and interpret average rate of change.

OpenStax: **Primary resources:** 3.3 Rates of Change and Behavior of Graphs

SUPPORTING SKILLS

- I Calculate average rate of change over an interval.
  SK-FUNC-AROC
- I Interpret average rate of change with contextual units.
  SK-FUNC-AROC-INTERPRET
- D Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

M21-FUN-007

I can compare functions represented by formulas, graphs, or tables.

OpenStax: **Primary resources:** Intermediate Algebra 2e 3.5 Relations and Functions
Supplemental resources: College Algebra 2e 3.1 Functions and Function Notation; College Algebra 2e 4.1 Linear Functions

SUPPORTING SKILLS

- I Compare functions shown in different representations.
  SK-FUNC-COMPARE-REP
- I Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH
- I Use tabular patterns to predict graph shape and behavior.
  SK-REP-TABLE-GRAPH

Polynomials and Factoring REQUIRED

OpenStax coverage: **Primary resources:** 5.1 Add and Subtract Polynomials; 5.3 Multiply Polynomials; 6.1 Greatest Common Factor and Factor by Grouping; 6.2 Factor Trinomials; 6.3 Factor Special Products; 6.5 Polynomial Equations

M21-POL-001

I can add and subtract polynomials.

OpenStax: **Primary resources:** 5.1 Add and Subtract Polynomials

SUPPORTING SKILLS

- A Identify and combine like terms.
  SK-ALG-LIKE-TERMS · inherited from middle school
- D Identify like polynomial terms by variable part and exponent.
  SK-POLY-LIKE-TERMS · first introduced in Math 12
- R Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12

M21-POL-002

I can multiply polynomials.

OpenStax: **Primary resources:** 5.3 Multiply Polynomials

SUPPORTING SKILLS

- A Apply the distributive property in either direction.
  SK-ALG-DISTRIBUTE · inherited from middle school
- D Multiply two binomials using distribution.
  SK-POLY-BINOMIAL-MULTIPLY · first introduced in Math 12
- D Multiply a monomial by a polynomial.
  SK-POLY-DISTRIBUTE · first introduced in Math 12
- A Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12

M21-POL-003

I can factor out a greatest common factor.

OpenStax: **Primary resources:** 6.1 Greatest Common Factor and Factor by Grouping

SUPPORTING SKILLS

- R Verify a factorization by multiplication.
  SK-FACTOR-CHECK · first introduced in Math 12
- D Factor the greatest common monomial factor from a polynomial.
  SK-FACTOR-GCF · first introduced in Math 12
- A Determine the numerical and variable greatest common factor of polynomial terms.
  SK-POLY-GCF · first introduced in Math 12
- A Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12

M21-POL-004

I can factor a quadratic trinomial.

OpenStax: **Primary resources:** 6.2 Factor Trinomials

SUPPORTING SKILLS

- A Verify a factorization by multiplication.
  SK-FACTOR-CHECK · first introduced in Math 12
- R Factor a quadratic trinomial with a nonunit leading coefficient.
  SK-FACTOR-TRINOMIAL-GENERAL · first introduced in Math 12
- R Factor a monic quadratic trinomial.
  SK-FACTOR-TRINOMIAL-MONIC · first introduced in Math 12
- A Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12

M21-POL-005

I can factor a difference of squares and other required special cases.

OpenStax: **Primary resources:** 6.3 Factor Special Products

SUPPORTING SKILLS

- A Verify a factorization by multiplication.
  SK-FACTOR-CHECK · first introduced in Math 12
- I Factor a difference of two squares.
  SK-FACTOR-DIFF-SQUARES
- A Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12

M21-POL-006

I can use factoring to determine the zeros of a polynomial.

OpenStax: **Primary resources:** 6.5 Polynomial Equations

SUPPORTING SKILLS

- A Factor a difference of two squares.
  SK-FACTOR-DIFF-SQUARES · first introduced in Math 21
- A Factor a quadratic trinomial with a nonunit leading coefficient.
  SK-FACTOR-TRINOMIAL-GENERAL · first introduced in Math 12
- A Factor a monic quadratic trinomial.
  SK-FACTOR-TRINOMIAL-MONIC · first introduced in Math 12
- I Apply the zero-product property.
  SK-FACTOR-ZERO-PRODUCT
- A Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12

Rational Expressions and Equations

REQUIRED

OpenStax coverage: **Primary resources:** 7.1 Multiply and Divide Rational Expressions; 7.2 Add and Subtract Rational Expressions; 7.4 Solve Rational Equations

M21-RAT-001

I can state excluded values for a rational expression.

OpenStax: **Primary resources:** 7.1 Multiply and Divide Rational Expressions

SUPPORTING SKILLS

- I — Determine excluded values from an original denominator.
  SK-RAT-RESTRICTIONS
- I — Recall and correctly use rational-expression terms including numerator, denominator, restriction, and excluded value.
  SK-RAT-VOCABULARY

M21-RAT-002

I can simplify a rational expression while preserving its restrictions.

OpenStax: **Primary resources:** 7.1 Multiply and Divide Rational Expressions

SUPPORTING SKILLS

- I — Cancel common factors rather than individual terms.
  SK-RAT-CANCEL-FACTORS
- I — Factor numerators and denominators completely.
  SK-RAT-FACTOR
- A — Determine excluded values from an original denominator.
  SK-RAT-RESTRICTIONS · first introduced in Math 21
- A — Recall and correctly use rational-expression terms including numerator, denominator, restriction, and excluded value.
  SK-RAT-VOCABULARY · first introduced in Math 21

M21-RAT-003

I can multiply and divide rational expressions.

OpenStax: **Primary resources:** 7.1 Multiply and Divide Rational Expressions

SUPPORTING SKILLS

- A — Add, subtract, multiply, and divide fractions.
  SK-NUM-FRACTION-OPERATE · inherited from middle school
- A — Cancel common factors rather than individual terms.
  SK-RAT-CANCEL-FACTORS · first introduced in Math 21
- A — Factor numerators and denominators completely.
  SK-RAT-FACTOR · first introduced in Math 21
- A — Determine excluded values from an original denominator.
  SK-RAT-RESTRICTIONS · first introduced in Math 21

M21-RAT-004

I can add and subtract rational expressions using a common denominator.

OpenStax: **Primary resources:** 7.2 Add and Subtract Rational Expressions

SUPPORTING SKILLS

- A Add, subtract, multiply, and divide fractions.
  SK-NUM-FRACTION-OPERATE · inherited from middle school
- I Combine rational expressions over a common denominator.
  SK-RAT-COMBINE
- I Determine the least common denominator of rational expressions.
  SK-RAT-LCD

M21-RAT-005

I can solve a rational equation and reject extraneous solutions.

OpenStax: **Primary resources:** 7.4 Solve Rational Equations

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I Clear denominators from a rational equation.
  SK-RAT-CLEAR-DENOMINATORS
- I Check candidate solutions against denominator restrictions.
  SK-RAT-EXTRANEOUS
- A Determine excluded values from an original denominator.
  SK-RAT-RESTRICTIONS · first introduced in Math 21

Exponents and Radicals REQUIRED

OpenStax coverage: **Primary resources:** Intermediate Algebra 2e 8.3 Simplify Rational Exponents; Intermediate Algebra 2e 8.2 Simplify Radical Expressions; Intermediate Algebra 2e 8.6 Solve Radical Equations
Supplemental resources: College Algebra 2e 2.6 Other Types of Equations

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M21-EXP-001

I can simplify expressions containing negative exponents.

OpenStax: **Primary resources:** 8.3 Simplify Rational Exponents

SUPPORTING SKILLS

- I Rewrite a negative exponent using a reciprocal.
  SK-EXP-NEGATIVE
- A Apply the product rule for exponents.
  SK-EXP-PRODUCT · first introduced in Math 12
- A Apply the quotient rule for exponents.
  SK-EXP-QUOTIENT · first introduced in Math 12

M21-EXP-002

I can rewrite expressions between radical and rational-exponent forms.

OpenStax: **Primary resources:** 8.3 Simplify Rational Exponents

SUPPORTING SKILLS

- I Interpret numerator and denominator roles in a rational exponent.
  SK-EXP-RATIONAL
- I Convert between radical and rational-exponent notation.
  SK-RAD-CONVERT
- I Read radical and rational-exponent notation and identify the index, radicand, base, numerator, and denominator.
  SK-RAD-NOTATION-READ

M21-EXP-003

I can simplify radical expressions.

OpenStax: **Primary resources:** 8.2 Simplify Radical Expressions

SUPPORTING SKILLS

- A Express a whole number as a product of prime factors.
  SK-NUM-PRIME-FACTOR · inherited from middle school
- I Extract perfect-power factors from a radical.
  SK-RAD-FACTOR-PERFECT

M21-EXP-004

I can apply exponent rules to expressions with integer and rational exponents.

OpenStax: **Primary resources:** 8.3 Simplify Rational Exponents

SUPPORTING SKILLS

- A Rewrite a negative exponent using a reciprocal.
  SK-EXP-NEGATIVE · first introduced in Math 21
- D Apply the power rule for exponents.
  SK-EXP-POWER · first introduced in Math 12
- D Apply the product rule for exponents.
  SK-EXP-PRODUCT · first introduced in Math 12
- D Apply the quotient rule for exponents.
  SK-EXP-QUOTIENT · first introduced in Math 12
- A Interpret numerator and denominator roles in a rational exponent.
  SK-EXP-RATIONAL · first introduced in Math 21
- I Apply the zero-exponent rule.
  SK-EXP-ZERO

M21-EXP-005

I can solve a basic equation involving a radical or rational exponent and check for extraneous solutions.

OpenStax: **Primary resources:** Intermediate Algebra 2e 8.6 Solve Radical Equations
Supplemental resources: College Algebra 2e 2.6 Other Types of Equations

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I Require an even-root radicand to be nonnegative when determining domain.
  SK-FUNC-DOMAIN-EVEN-ROOT
- I Check a candidate solution for extraneous results.
  SK-RAD-CHECK
- I Isolate a radical expression before solving.
  SK-RAD-ISOLATE

Quadratic Functions and Equations REQUIRED

OpenStax coverage: **Primary resources:** 9.6 Graph Quadratic Functions Using Properties; 9.7 Graph Quadratic Functions Using Transformations; 6.5 Polynomial Equations; 9.1 Solve Quadratic Equations Using the Square Root Property; 9.2 Solve Quadratic Equations by Completing the Square; 9.3 Solve Quadratic Equations Using the Quadratic Formula; 9.5 Solve Applications of Quadratic Equations
Supplemental resources: 9.6 Graph Quadratic Functions Using Properties; 6.5 Polynomial Equations

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M21-QUAD-001

I can identify key features of a quadratic function from its graph or equation.

OpenStax: **Primary resources:** 9.6 Graph Quadratic Functions Using Properties

SUPPORTING SKILLS

- I — Identify standard, factored, and vertex forms.
  SK-QUAD-FORM-IDENTIFY
- I — Determine x- and y-intercepts of a quadratic function.
  SK-QUAD-INTERCEPTS
- I — Determine the vertex and axis of symmetry.
  SK-QUAD-VERTEX
- I — Recall and correctly use quadratic-function terms including parabola, vertex, axis of symmetry, intercept, zero, and solution.
  SK-QUAD-VOCABULARY

M21-QUAD-002

I can connect standard, factored, and vertex forms to features of a parabola.

OpenStax: **Primary resources:** 9.7 Graph Quadratic Functions Using Transformations
Supplemental resources: 9.6 Graph Quadratic Functions Using Properties; 6.5 Polynomial Equations

SUPPORTING SKILLS

- I — Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method.
  SK-QUAD-FORM-CONVERT
- D — Identify standard, factored, and vertex forms.
  SK-QUAD-FORM-IDENTIFY · first introduced in Math 21
- A — Determine x- and y-intercepts of a quadratic function.
  SK-QUAD-INTERCEPTS · first introduced in Math 21
- A — Determine the vertex and axis of symmetry.
  SK-QUAD-VERTEX · first introduced in Math 21
- D — Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21

M21-QUAD-003

I can solve a quadratic equation by factoring.

OpenStax: **Primary resources:** 6.5 Polynomial Equations

SUPPORTING SKILLS

- A — Factor a quadratic trinomial with a nonunit leading coefficient.
  SK-FACTOR-TRINOMIAL-GENERAL · first introduced in Math 12
- A — Factor a monic quadratic trinomial.
  SK-FACTOR-TRINOMIAL-MONIC · first introduced in Math 12
- D — Apply the zero-product property.
  SK-FACTOR-ZERO-PRODUCT · first introduced in Math 21
- I — Determine real and complex zeros using an appropriate method.
  SK-POLY-ZEROS

M21-QUAD-004

I can solve a quadratic equation using the square-root property.

OpenStax: **Primary resources:** 9.1 Solve Quadratic Equations Using the Square Root Property

SUPPORTING SKILLS

- I — Apply the square-root property with both signs.
  SK-QUAD-SQUARE-ROOT
- A — Extract perfect-power factors from a radical.
  SK-RAD-FACTOR-PERFECT · first introduced in Math 21

M21-QUAD-005

I can solve a quadratic equation by completing the square.

OpenStax: **Primary resources:** 9.2 Solve Quadratic Equations by Completing the Square

SUPPORTING SKILLS

- A — Simplify an algebraic expression using operation properties.
  SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I — Create a perfect-square trinomial by completing the square.
  SK-QUAD-COMPLETE-SQUARE

M21-QUAD-006

I can solve a quadratic equation using the quadratic formula.

OpenStax: **Primary resources:** 9.3 Solve Quadratic Equations Using the Quadratic Formula

SUPPORTING SKILLS

- I — Use the discriminant to predict the number and type of solutions.
  SK-QUAD-DISCRIMINANT
- I — Substitute coefficients accurately into the quadratic formula.
  SK-QUAD-FORMULA
- A — Extract perfect-power factors from a radical.
  SK-RAD-FACTOR-PERFECT · first introduced in Math 21

M21-QUAD-007

I can choose an efficient method for solving a quadratic equation.

OpenStax: **Primary resources:** 9.3 Solve Quadratic Equations Using the Quadratic Formula

SUPPORTING SKILLS

- A — Apply the zero-product property.
  SK-FACTOR-ZERO-PRODUCT · first introduced in Math 21
- A — Create a perfect-square trinomial by completing the square.
  SK-QUAD-COMPLETE-SQUARE · first introduced in Math 21
- A — Substitute coefficients accurately into the quadratic formula.
  SK-QUAD-FORMULA · first introduced in Math 21
- I — Select an efficient quadratic-solving method from equation structure.
  SK-QUAD-METHOD-SELECT
- A — Apply the square-root property with both signs.
  SK-QUAD-SQUARE-ROOT · first introduced in Math 21

M21-QUAD-008

I can graph a quadratic function using its vertex, intercepts, and symmetry.

OpenStax: **Primary resources:** 9.6 Graph Quadratic Functions Using Properties

SUPPORTING SKILLS

- A Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method.
  SK-QUAD-FORM-CONVERT · first introduced in Math 21
- D Determine x- and y-intercepts of a quadratic function.
  SK-QUAD-INTERCEPTS · first introduced in Math 21
- D Determine the vertex and axis of symmetry.
  SK-QUAD-VERTEX · first introduced in Math 21
- D Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21

M21-QUAD-009

I can create and interpret a quadratic model for a projectile, area, or optimization problem.

OpenStax: **Primary resources:** 9.5 Solve Applications of Quadratic Equations

SUPPORTING SKILLS

- D Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A Restrict a model to meaningful contextual inputs.
  SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- I Interpret vertex and zeros in an application.
  SK-QUAD-MODEL-FEATURES
- A Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

Extension Content

EXTENSION

OpenStax coverage: **Primary resources:** 8.8 Use the Complex Number System; 9.3 Solve Quadratic Equations Using the Quadratic Formula

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M21-EXT-001

I can perform basic arithmetic with complex numbers.

OpenStax: **Primary resources:** 8.8 Use the Complex Number System

SUPPORTING SKILLS

- I — Simplify integer powers of the imaginary unit.
  SK-CPLX-I-POWERS
- I — Add, subtract, and multiply complex numbers.
  SK-CPLX-OPERATE

M21-EXT-002

I can express the nonreal solutions of a quadratic equation using complex numbers.

OpenStax: **Primary resources:** 9.3 Solve Quadratic Equations Using the Quadratic Formula

SUPPORTING SKILLS

- A — Simplify integer powers of the imaginary unit.
  SK-CPLX-I-POWERS · first introduced in Math 21
- I — Express a quadratic solution with a negative discriminant in complex form.
  SK-CPLX-QUAD-SOLUTION
- A — Use the discriminant to predict the number and type of solutions.
  SK-QUAD-DISCRIMINANT · first introduced in Math 21